

ADITYA JOSHI

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EDUCATION

Savitribai Phule Pune University

Nov 2022 – May 2026

BE in Computer Engineering with Honours in DS and ML

- CGPA: 7.9/10

EXPERIENCE

AI/ML Intern — Hilwitz Solutions Pvt. Ltd.

Nov 2025 – Apr 2026

- Completed a six-month internship focused on AI/ML development, building and deploying machine learning solutions.

Intern – GenAI — Maharashtra Knowledge Corporation Ltd. (MKCL)

Mar 2025 – Sept 2025

- Built a Flask-based API and Vue.js frontend for an Employee Directory System with authentication and role-based access.
- Developed a multilingual Graph RAG chatbot running on a locally hosted LLM with Neo4j, capable of answering queries on government schemes datasets in both English and Marathi.
- Strengthened expertise in GenAI applications, multilingual NLP, knowledge graphs, and full-stack development.

Backend Development Workshop — Netcamp Solutions Pvt. Ltd.

Jun 2024 – Aug 2024

- Developed and optimized RESTful APIs using Python (Flask) to enhance system efficiency.
- Designed and implemented an SQL database schema to ensure smooth data management and retrieval.
- Worked on a JavaScript-based front-end project, improving UI/UX in collaboration with peers.

PUBLICATIONS

Understanding Quantum Computers

Jan 2024

Aditya Joshi, Pratham Dixit, Avdhut Pithe — MMCOE, Yearly Magazine

Generated a truly random number using a quantum gate and verified how regular logic gates cannot create truly random numbers.

Explained the difference between quantum and regular computers.

Conversational Chatbot for Government Schemes Using Neo4j Knowledge Graph and Multilingual Voice Interface

Aditya Joshi, Anshul Dhore, Girija Dahiphale, Pratham Dixit, Snehal Karad

GraphRAG-powered chatbot for Indian government schemes, combining a Neo4j knowledge graph with a locally deployed LLM and a trilingual (English/Hindi/Marathi) voice interface. 92.4% intent accuracy, 95.2% Top-3 retrieval accuracy across 512 evaluated queries.

PROJECTS

SalesCope — Sales Analytics SaaS with AI Insights

github.com/adi-21-4/salescope

- Built a multi-tenant SaaS (FastAPI, PostgreSQL, React + TypeScript) that turns an uploaded sales CSV/Excel file into an auto-generated dashboard, AI-written insights, and revenue forecasts, with tenant isolation scoped by shop ID.
- Designed a heuristic column-mapping pipeline with confidence scoring and upload deduplication via content hashing, and integrated AI through a swappable provider to translate questions into a constrained query DSL over deterministic Pandas logic.

Government Schemes Graph RAG Chatbot

github.com/adi-21-4/graphbased-chatbot

- Built a full-stack, voice-enabled Retrieval-Augmented Generation chatbot answering government scheme queries in English and Marathi.
- Used a Neo4j knowledge graph with a local LLM to translate natural-language questions into Cypher queries for relationship-based retrieval.
- Implemented speech-to-text via the Web Speech API and offline text-to-speech with pyttsx3; frontend in Vue 3 + Tailwind CSS, backend in Python (Flask).

AI Powered Chat-bot

github.com/adi-21-4/chatbot

- Developed an AI-driven chatbot to handle basic student queries related to faculty and administration.
- Used JavaScript for the frontend, with a JSON-based database for storing responses.
- Integrated tokenization to improve query understanding and accuracy.

Flask API Backend

github.com/adi-21-4/api-backend-with-flask

- Built a Flask-based REST API implementing CRUD operations, file upload, and retrieval against a MySQL database.
- Organized the codebase into controller and model layers, with authentication handling and role based access.

Movie Recommendation System

github.com/adi-21-4/movie-recommendation-system

- Built a machine learning-based recommendation system using Python (Pandas, Scikit-Learn).
- Utilized collaborative filtering algorithms to suggest movies based on user preferences.
- Designed an interactive web interface using Flask and HTML/CSS.

TECHNOLOGIES

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, Kotlin

Technologies: .NET, Microsoft SQL Server, PostgreSQL, MySQL, Neo4j, LangChain, Claude API, FastAPI, Flask, React, Vue.js, Docker, Scikit-Learn, Postman